Statistical Graphics

Objective

- Bar Plot
- Histogram
- Scatter Plot
- Density Plot
- Box Plot
- ggplot2

Bar Plot

- Bar chart can be drawn with the help of function **barplot**()

Syntax :

barplot(height, horiz = FALSE, main = NULL, sub = NULL, xlab = NULL, ylab = NULL,...)

Where

height : A vector / matrix

horiz : logical; If TRUE then graph results in horizontal bar graph

main : Title of Graph

xlab : X-axis label

ylab : Y-axis label

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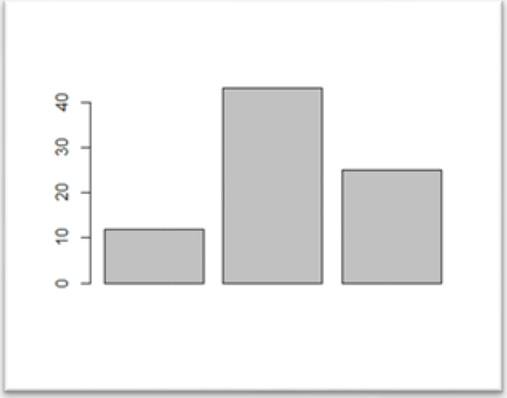
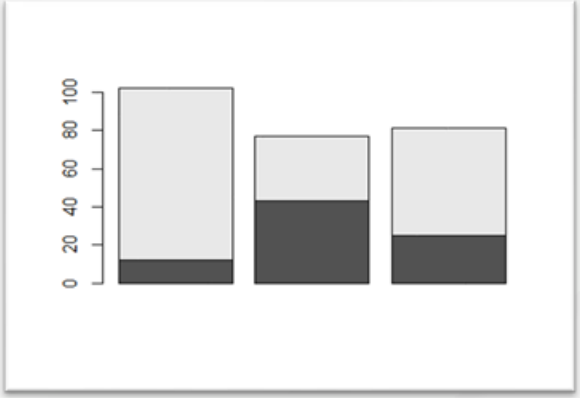
Bar Plot Examples

```
> ProductA <- c(12,43,25)
> barplot(ProductA)
```




```
> ProductA <- c(12,43,25)
> ProductB <- c(90,34,56)
> rb <- rbind(ProductA,ProductB)
> barplot(rb)
```



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Bar Plot Examples

```
> barplot(table(Gender),col = "brown",main = "Gender-Wise Distribution")
```

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Histogram

- Bar Plot on binned data can be said to be a histogram
- Difference between histogram and bar plot is that in bar plot we are free to rearrange the bars whereas in histogram we are not
- We can know the distribution of any variable with histogram

Syntax :

hist(x, breaks = "", include.lowest = TRUE, main = paste("Histogram of" , xname), xlim = range(breaks), xlab = xname, ylab, ...)

Where x : numeric vector or a function which can generate a numeric vector

breaks : specifying the number of breaks or a numeric vector of breaks

xlim : Scale of limits on X axis

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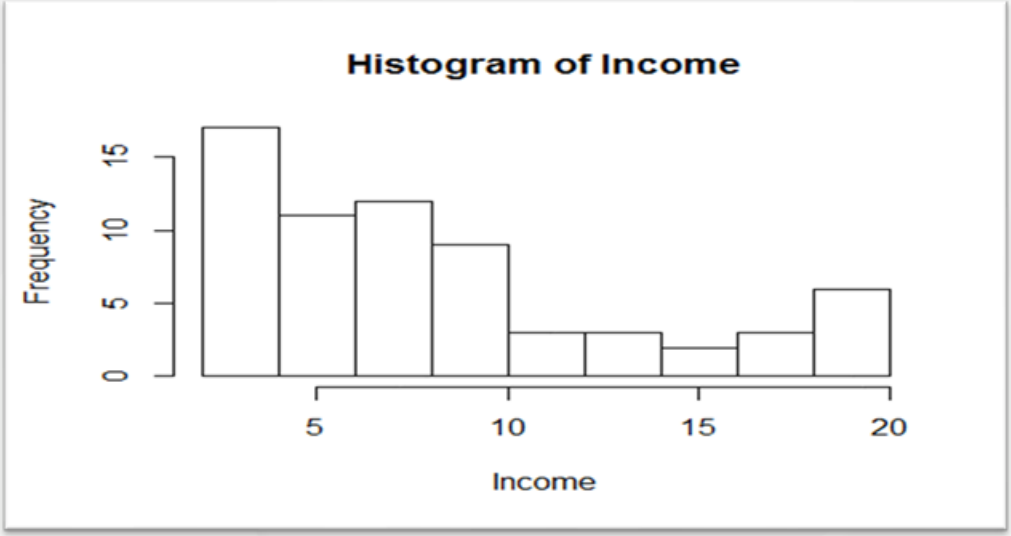
Histogram Examples

```
> hist(Income)
```



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Scatter Plot

- Scatter Plot can be understood as plot of points on 2-dimensional space of XY-plane
- Scatter plot gives us the information about correlation between the two variables plotted

- Syntax :
  - plot(x, y, type, col, pch...)
  - Where x , y : numeric vector or a function generating numerical vector
  - type : p for points, l for lines, b for both, s for stair
  - col : Colour of plotting character
  - pch : Plotting character

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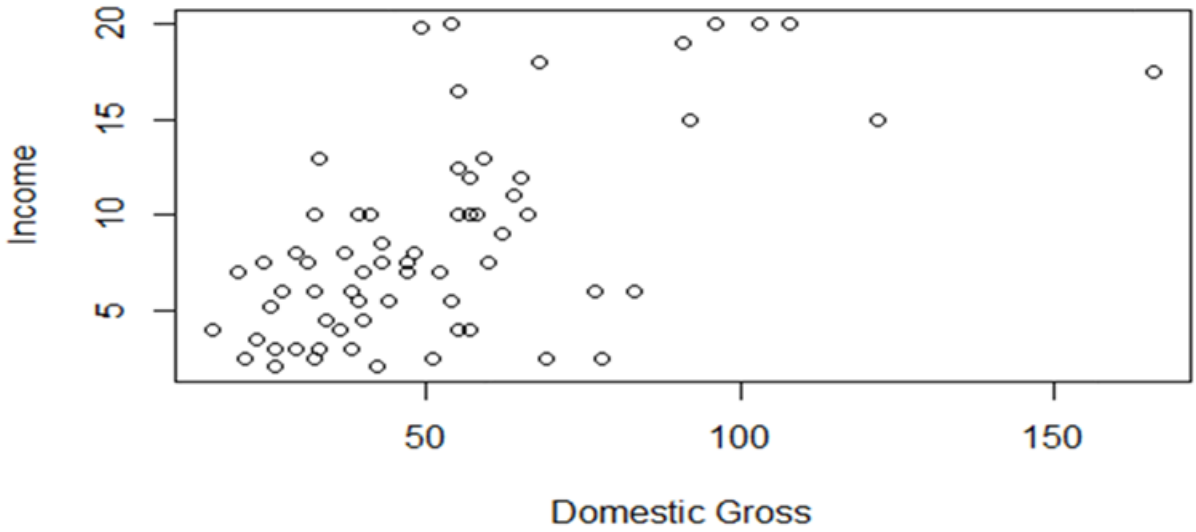
Scatter Plot Examples

```
> plot(Domestic.Gross,Income,xlab="Domestic Gross",ylab="Income")
```



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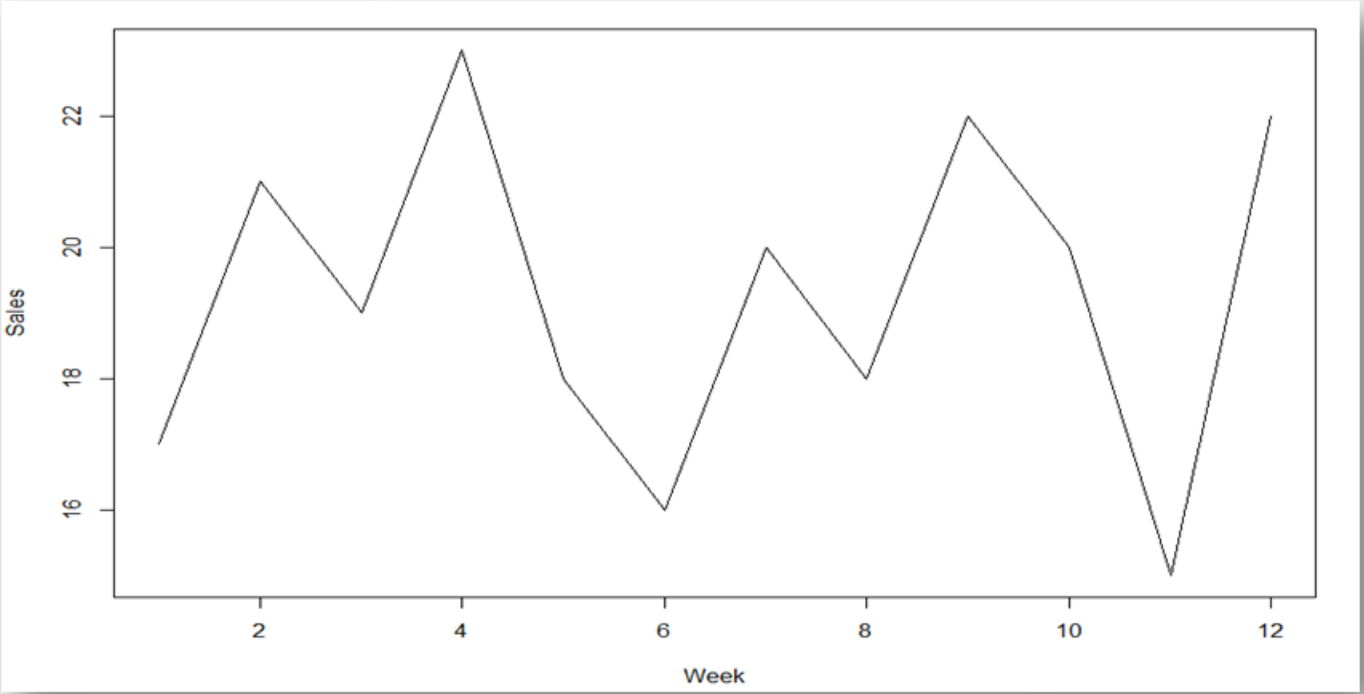
Line Plot using plot()

```
plot(gasoline$week,gasoline$sales,xlab = "week",ylab = "sales",type = 'l')
```



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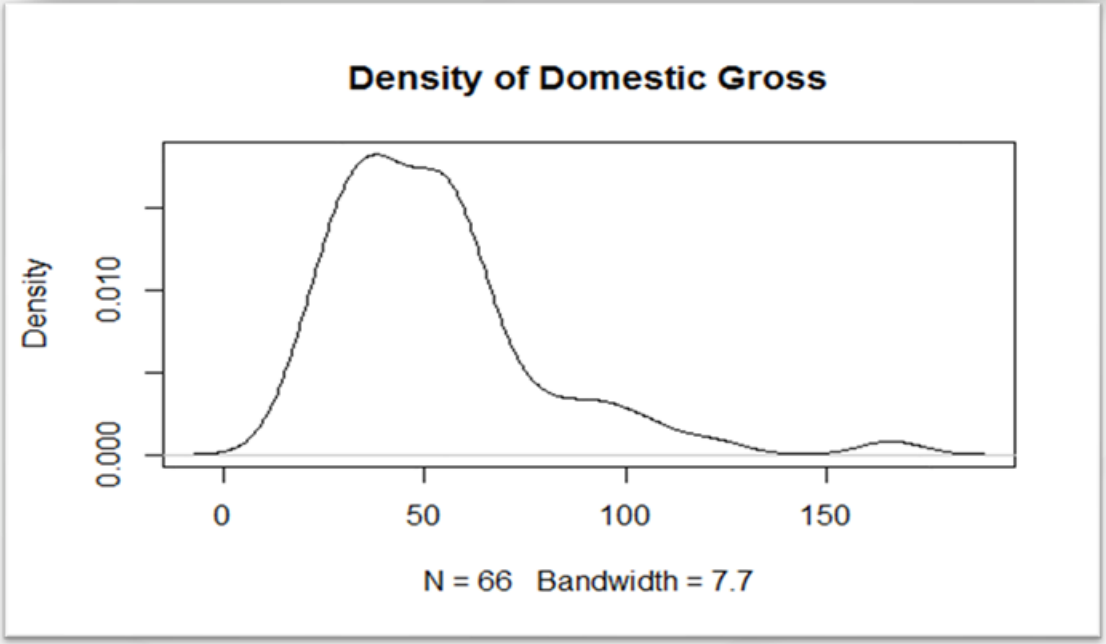
Density Plot

- Density plot is smoothened form of histogram
- Density plot can be plotted with function plot() by calling density function

```
> plot(density(Domestic.Gross),main="Density of Domestic Gross")
```



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Boxplot

- Before we understand box plot we need to understand the quartiles
- Quartiles divide the given data into four equal parts.

- **Data represented as a line**

- Inter-quartile range (IQR) is given by the formula:
- IQR= $Q3-Q1$

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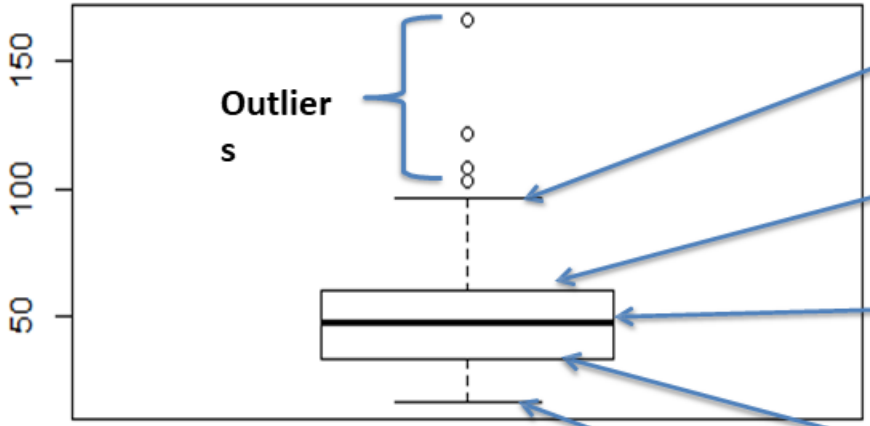
Describing with Box Plots



Maximum value excluding outliers

Quartile 3

Quartile 2 / Median

Quartile 1

Minimum value excluding outliers

Here Outliers are the observations which are 1.5*IQR from the edge of the box

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Boxplot

- Boxplot gives us the spread of the data
- Syntax : boxplot(x)
- Where x : numeric vector OR
- boxplot(y ~ x)
- Where y : numeric vector and x : factor

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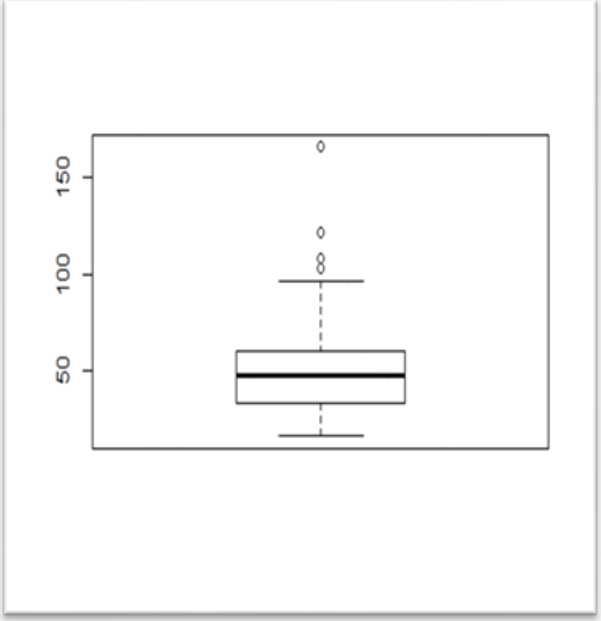
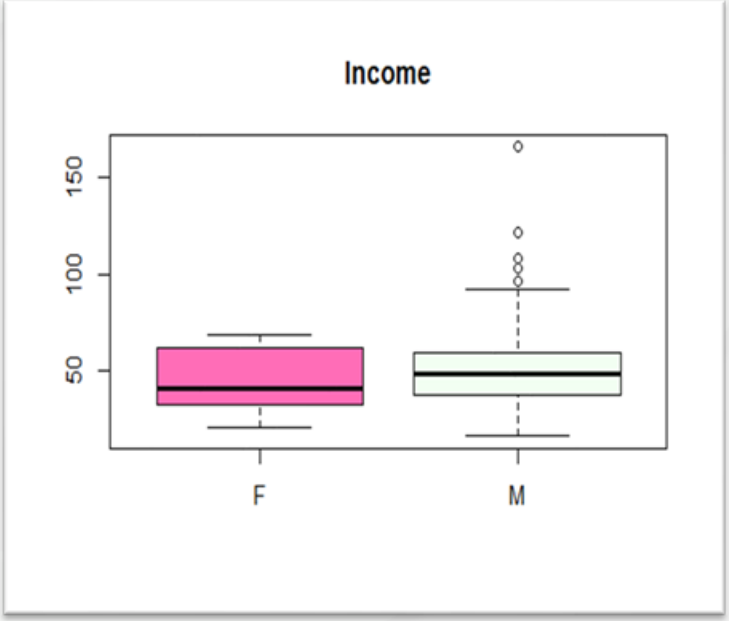
Boxplot Examples

```
> boxplot(Domestic.Gross)
```




```
> boxplot(Domestic.Gross ~ Gender,
+         col = c("hotpink","honeydew1"),main="Income")
```



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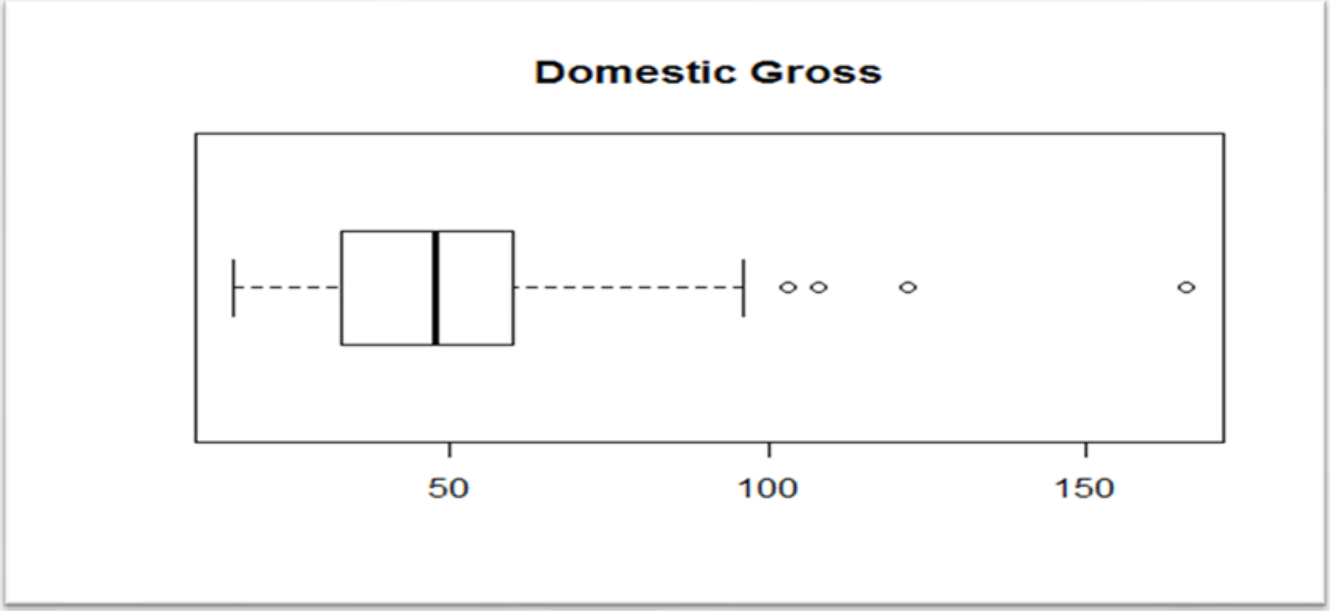
Boxplot Examples

```
> boxplot(Domestic.Gross,horizontal = TRUE,main="Domestic Gross")
```



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Enhanced Graphics Package

About ggplot2

- Package ggplot2 provides a method of creating innovative graphs based on graphical grammar
- There are four graphic systems in R currently.

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Four Graphics Systems in R

1. The baseGraphics System written by the Ross Ihaka included by default in every R installation
2. The gridgraphics system written by Paul Murrell (2011)
3. The latticegraphics system written by Deepayan Sarkar (2008)
4. The ggplot2graphics system written by Hadley Wickham (2009)

Graphics

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Base Graphics

- We have already covered it in the previous sessions
- Composed of functions like plot(), boxplot(), barplot() etc.

Grid Graphics

- Implemented by package grid offers a low level alternative to the standard graphics system. But the gridpackage doesn't provide functions for producing statistical graphics or complete plots.

Lattice Graphics

- Implements trellis graphs with package lattice provides a comprehensive system for creating statistical graphics built using package grid

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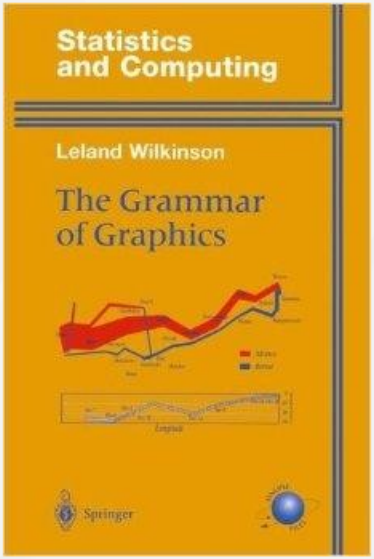
Ggplot2 Graphics

- The package *ggplot2* has been written by Hadley Wickham
- Provides a system for creating graphs based on the grammar of graphics described by Wilkinson and expanded by Wickham



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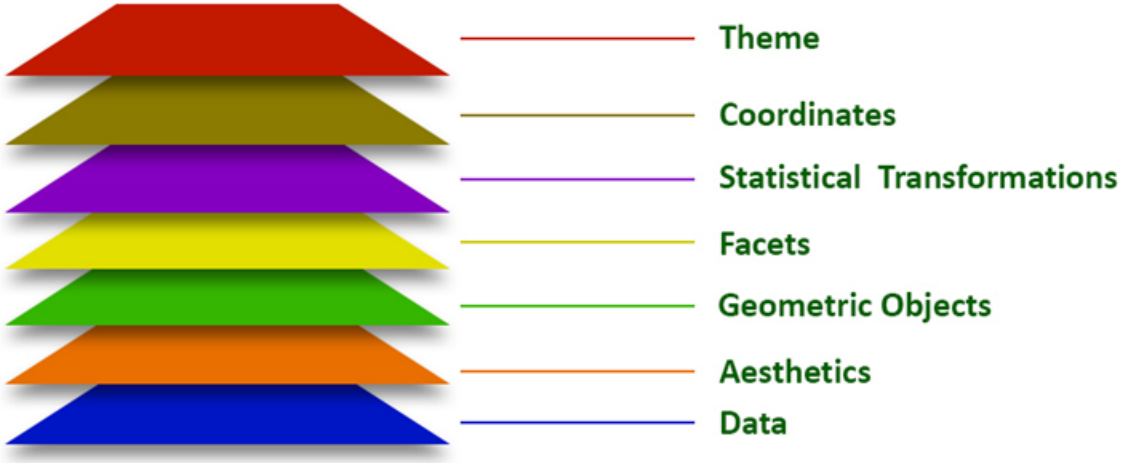
Main Components of the Grammar of Graphics



Basic Components of ggplot()

- **Data**: The dataset that you want to visualize. It is passed to the ggplot() function.
- **Aesthetics (aes)**: Defines the mapping between the variables in your data and the visual properties of the plot, such as x and y coordinates, colors, shapes, and sizes.
- **Geometries (geoms)**: These represent the actual shapes or objects that you want to plot, such as points, lines, bars, etc. Common geometries include geom_point(), geom_line(), geom_bar(), and geom_histogram().
- **Facets**: These allow you to split your data into subsets and create multiple plots, one for each subset. This is done using facet_wrap() or facet_grid().

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Basic Components of ggplot()…..

- **Statistical transformations (stats)**: These are built-in transformations that you can apply to your data before plotting, like binning in histograms or smoothing in line plots.
- **Coordinates**: These define the coordinate system used in the plot. The default is Cartesian coordinates, but you can also use polar coordinates (coord_polar()) or others.
- **Themes**: These control the overall look of the plot, including text, background, and grid lines. You can use pre-built themes like theme_minimal() or customize your own.

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Building Plots in ggplot()

- Plots are built up in layers
  - Plotting the data
  - Overlaying the summary
  - Annotating the graph

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Plotting by ggplot()

- The main function in ggplot2 is ggplot(), which creates a plot object to which layers (like points, lines, bars) can be added using the + operator.
- **ggplot(data, aes(x, y)) + geom_layer() + other_layers_or_customizations**
  - **data**: The data frame containing the data to be plotted.
  - **aes()**: Defines the aesthetic mappings (e.g., x, y, color, size).
  - **geom_layer()**: Specifies the type of plot (e.g., points, bars, lines).
  - Additional layers like scales, facets, and themes can be added to customize the visualization.

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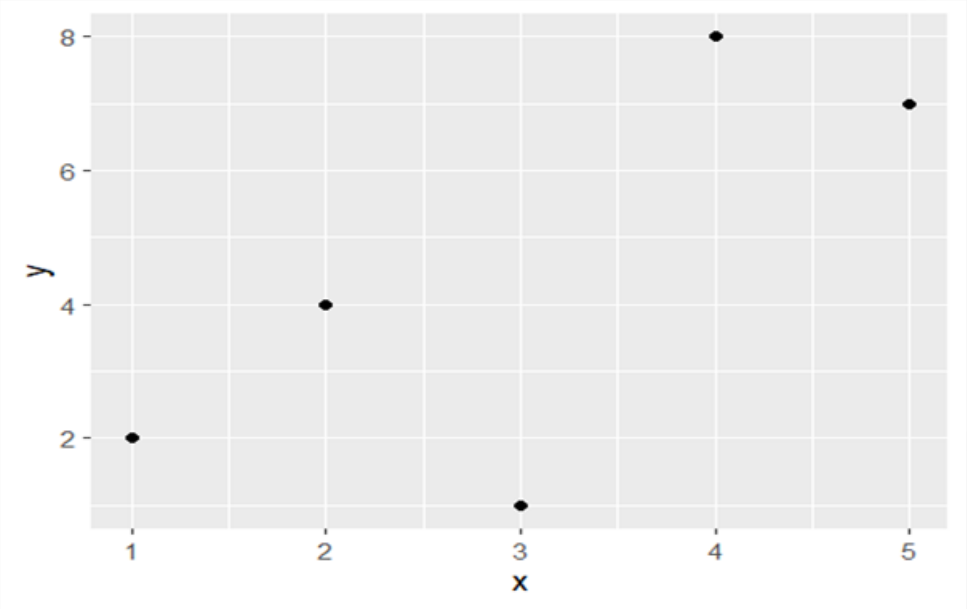
geom_*() functions

Geom	Graph Type
geom_point()	Scatter Plot
geom_line()	Line Graph
geom_histogram()	Histogram
geom_density()	Density Plot
geom_smooth()	Regression Line
geom_boxplot()	Boxplot
geom_bar()	Bar PLot

Scatter Plot

```
> library(ggplot2)
>
> data <- data.frame(  x = c(1, 2, 3, 4, 5),
+                      y = c(2, 4, 1, 8, 7),
+                      group = c("A", "B", "A", "B",
"A"))
> ggplot(data, aes(x, y)) + geom_point()
```



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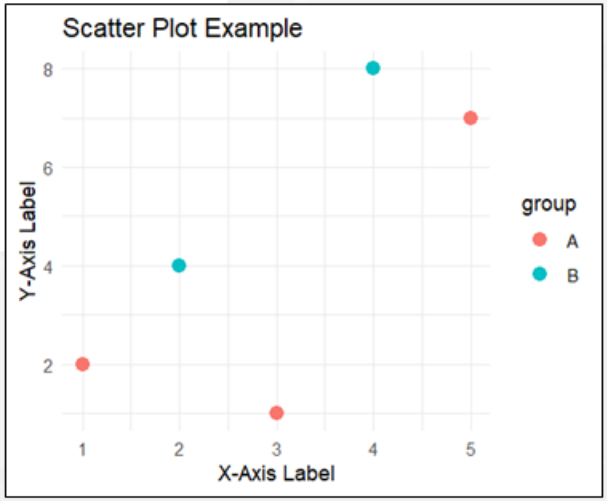
Scatter Plot

- Scatter plot can be generated with geom_point() function

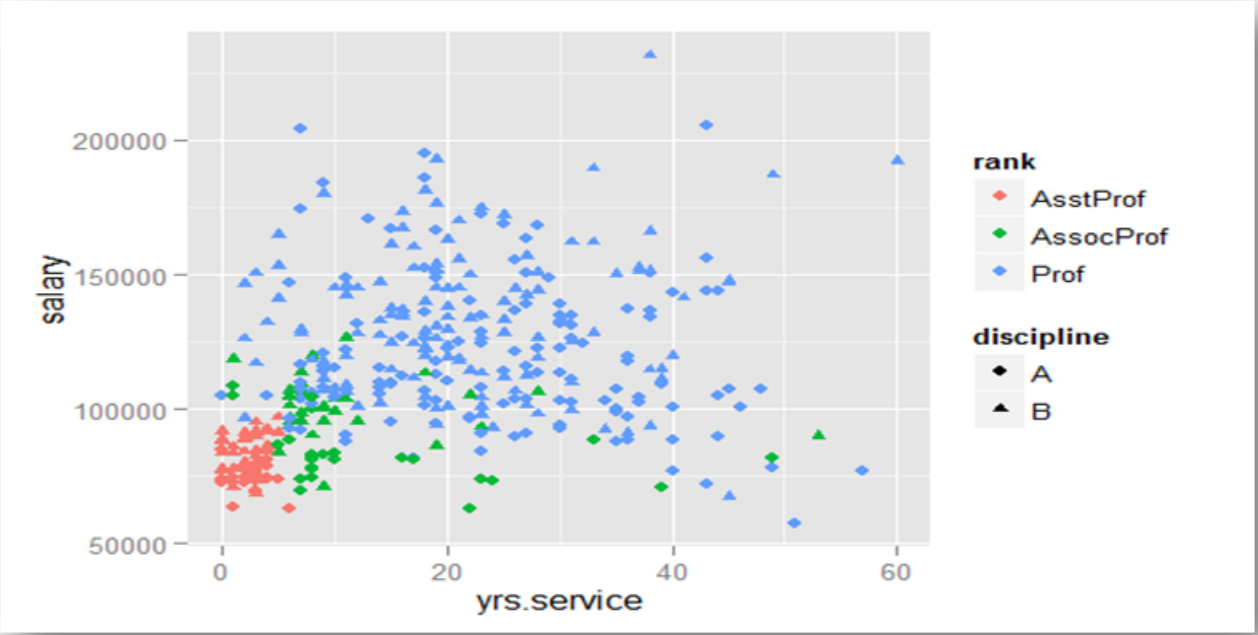
```
> # Example dataset
> data <- data.frame(
+   x = c(1, 2, 3, 4, 5),
+   y = c(2, 4, 1, 8, 7),
+   group = c("A", "B", "A", "B", "A")
+ )
>
> # Create a scatter plot
> ggplot(data, aes(x = x, y = y, color = group)) +
+   geom_point(size = 3) +
+   labs(
+     title = "Scatter Plot Example",
+     x = "X-Axis Label",
+     y = "Y-Axis Label"
+   ) +
+   theme_minimal()
```



Grouping in Scatter Plot

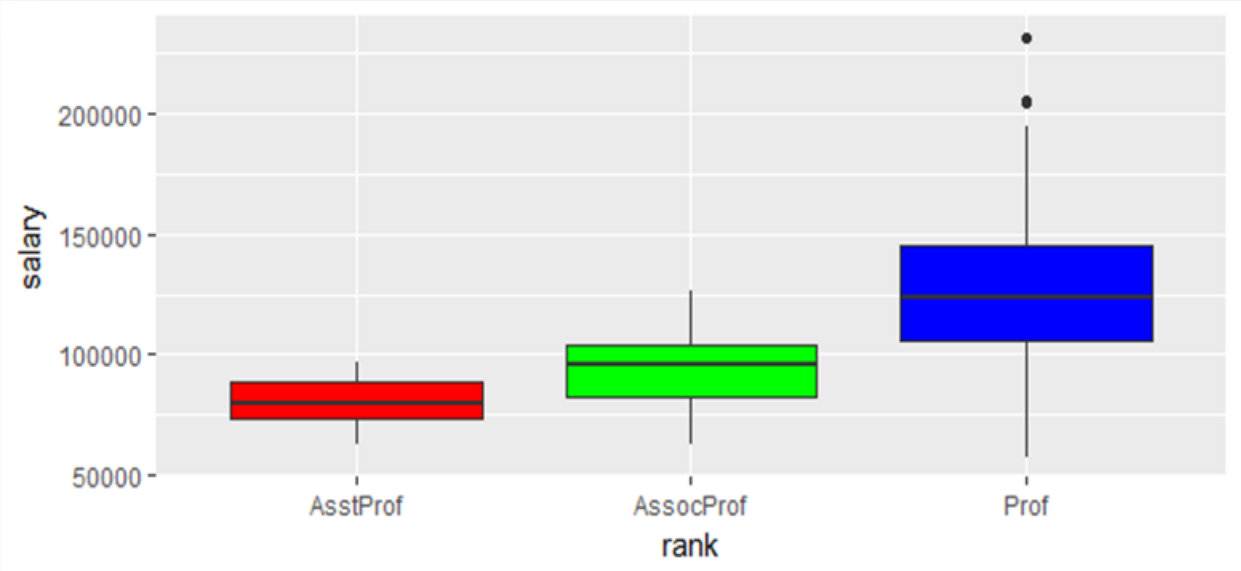
```
install.packages("carData") # Load the package
library(carData) # View the Salaries dataset
data(Salaries)
```

```
ggplot(Salaries,aes(yrs.service,salary,color=rank,shape=discipline))+
 geom_point()
```



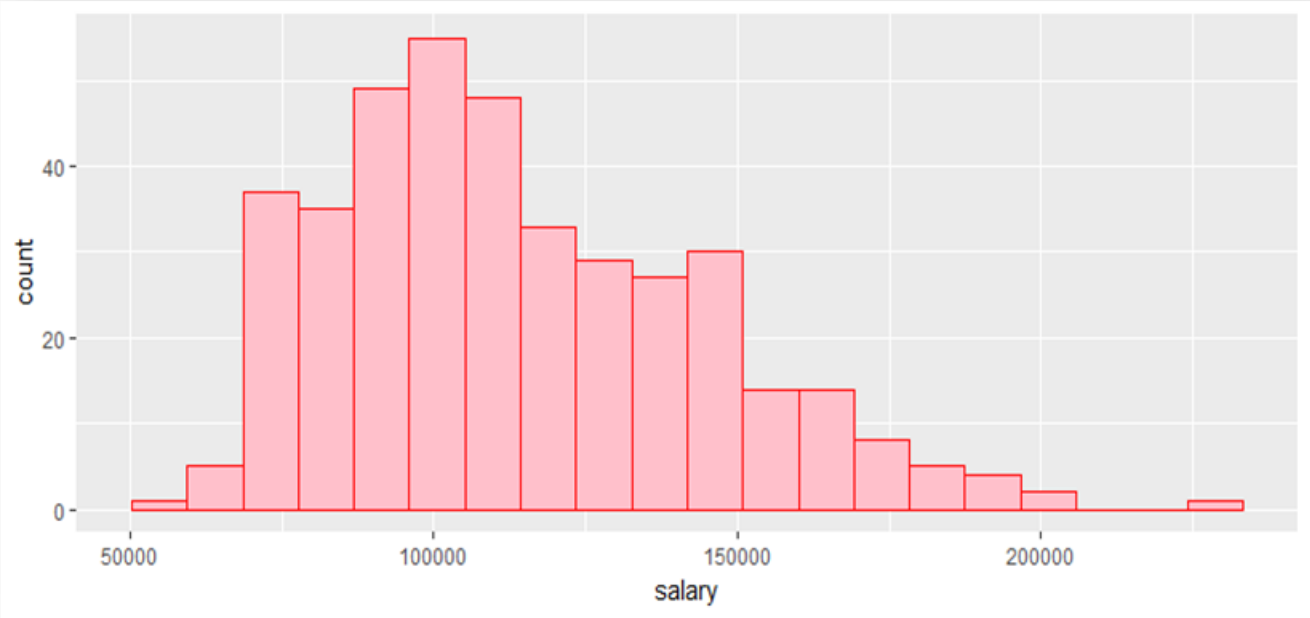
Boxplot

```
ggplot(Salaries, aes(x=rank,y=salary))+
  geom_boxplot(fill=c("red","green","blue"))
```



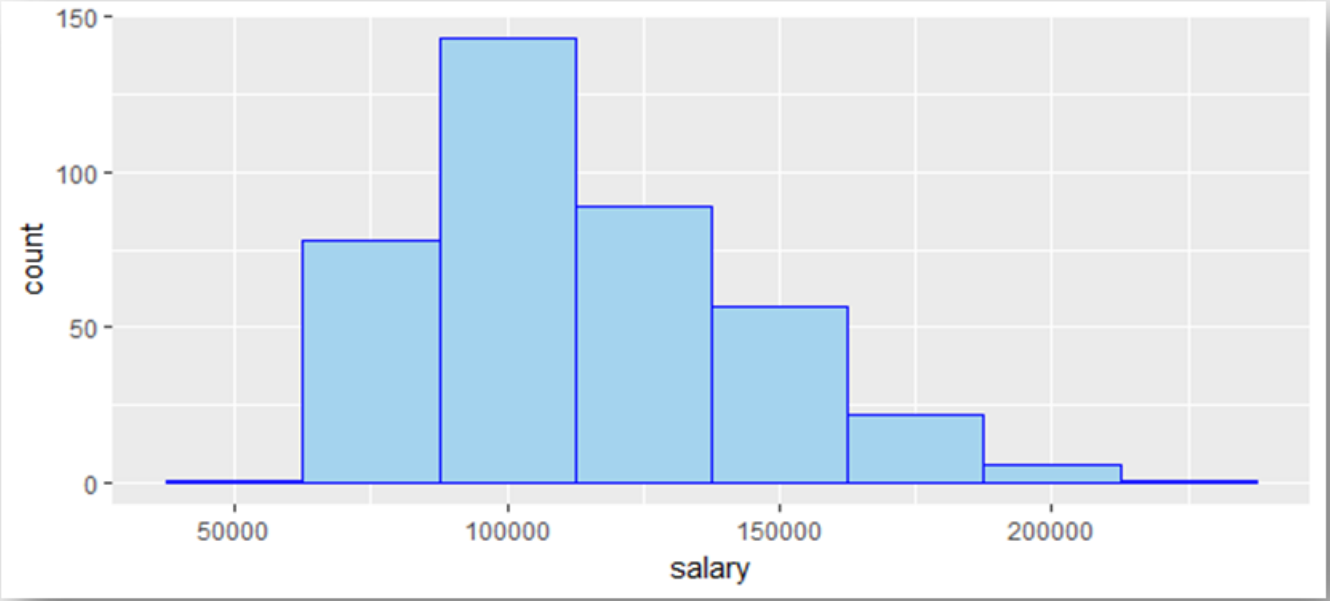
Histogram

```
ggplot(Salaries,aes(x=salary)) +
  geom_histogram(bins=20,fill="pink",color="red")
```



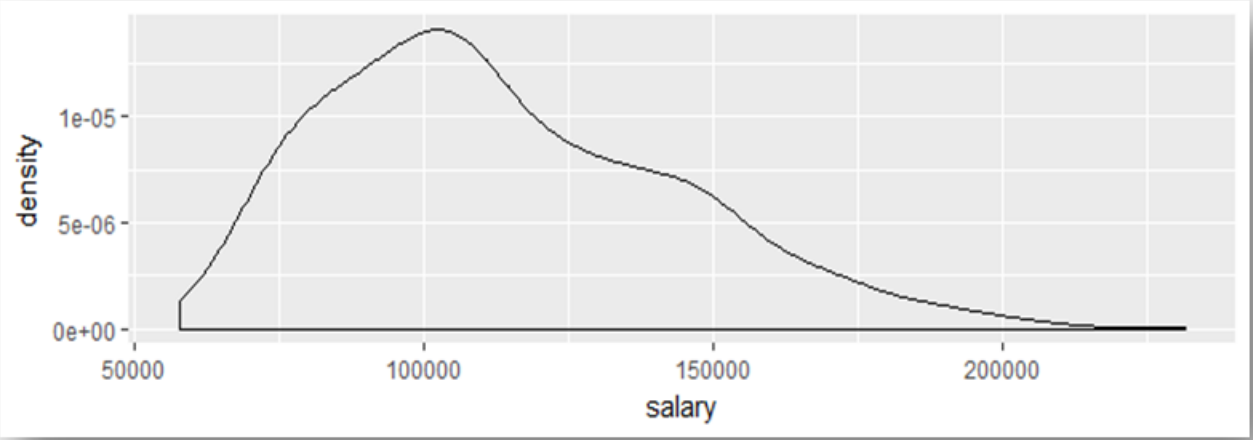
Histogram

```
ggplot(Salaries,aes(x=salary)) +
  geom_histogram(binwidth = 25000,fill="lightskyblue2",color="blue")
```



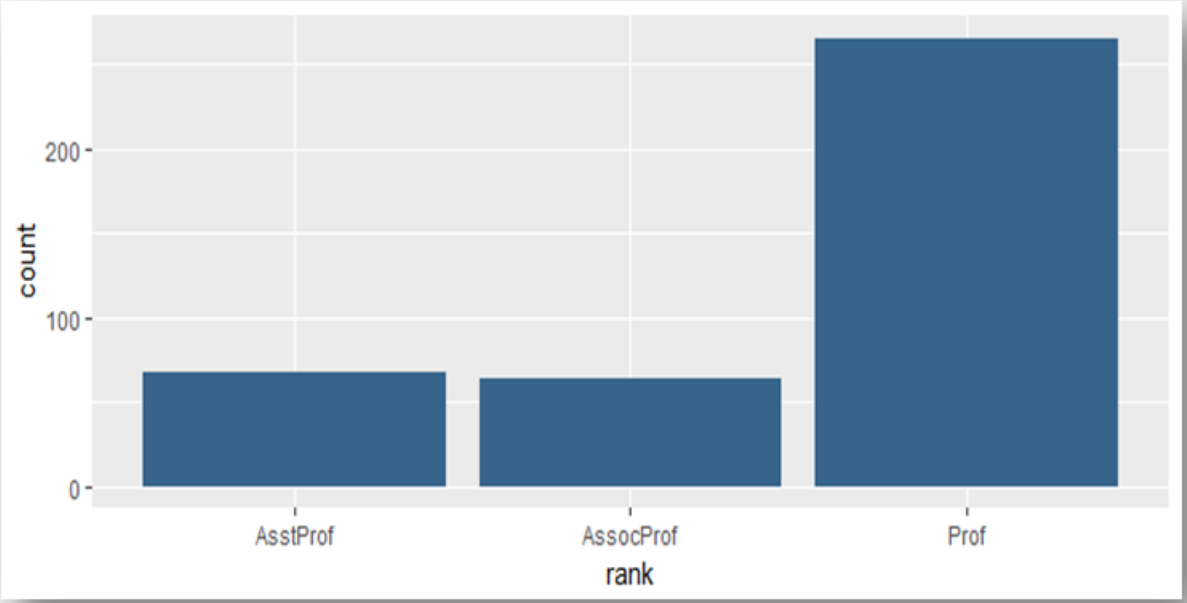
Density Plot

```
ggplot(Salaries, aes(x=salary))+geom_density()
```



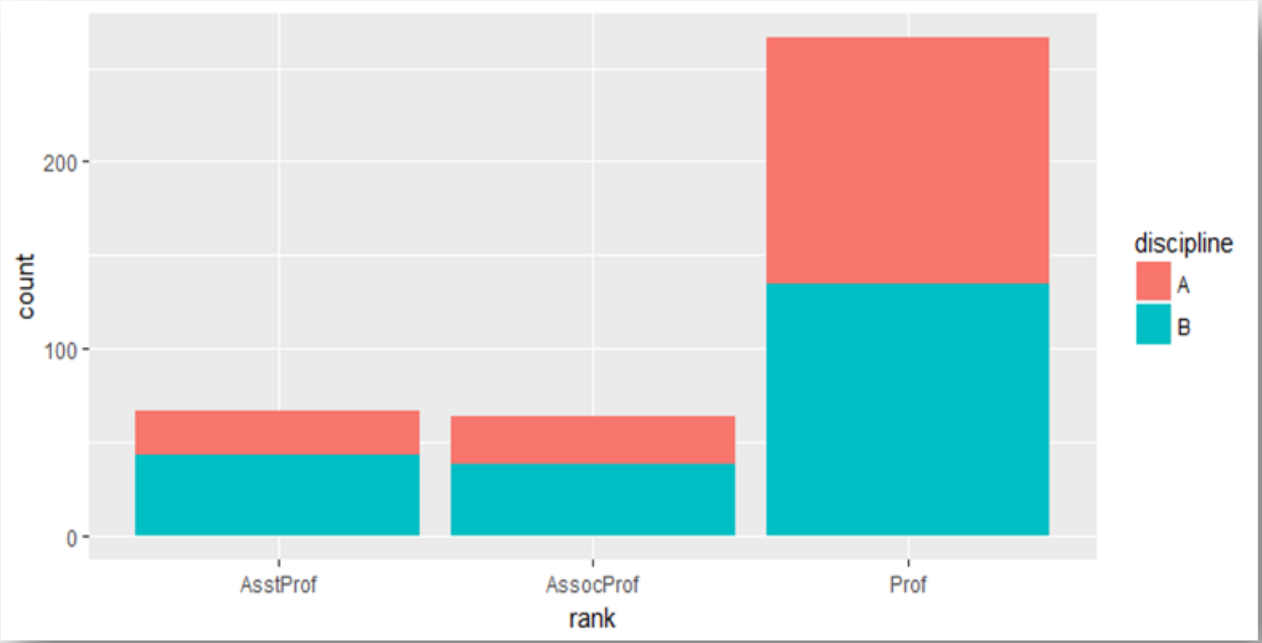
Barplot

```
ggplot(Salaries, aes(rank))+geom_bar(fill="steelblue4")
```



Barplot

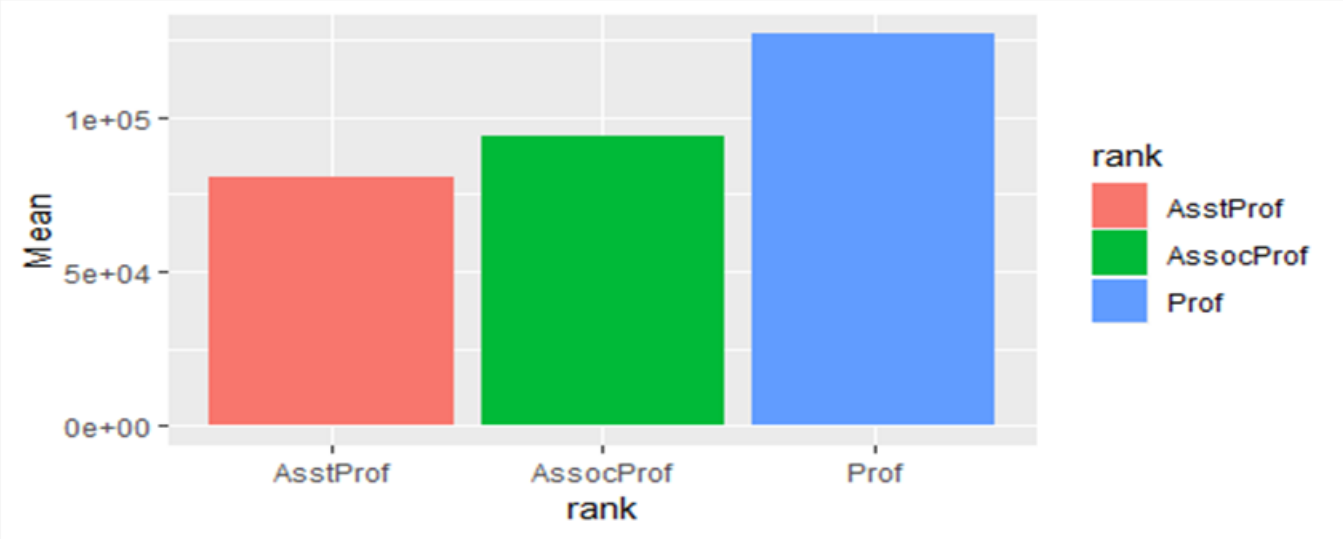
```
ggplot(Salaries, aes(rank,fill=discipline))+geom_bar()
```



Plotting Summarized Values

```
meanSals <- Salaries %>%
                group_by(rank) %>%
                summarise(Mean=mean(salary,na.rm = T))
```

```
ggplot(data=meanSals,aes(x=rank,y=Mean,fill=rank))+
  geom_bar(stat = "identity")
```



Facets

- Sometimes, the relationship become clearer if the graphs are shown side by side
- These are also called trellis graphs

Syntax	Effect
facet_grid(rowvar ~ colvar)	Separate plots for each combination of rowvar and colvar in grid form
facet_grid(rowvar ~ .)	Separate plots of each category in rowvar into a single column
facet_grid(. ~ colvar)	Separate plots of each category in colvar into a single row

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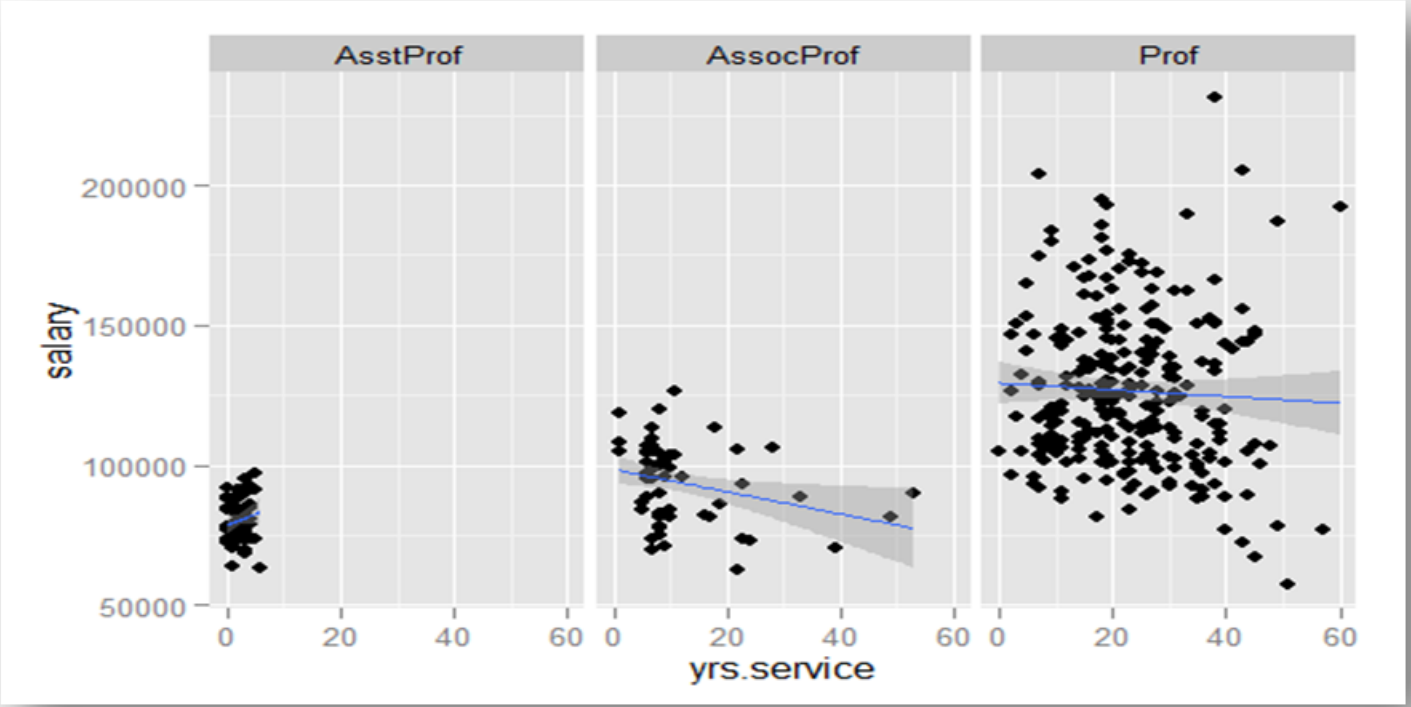
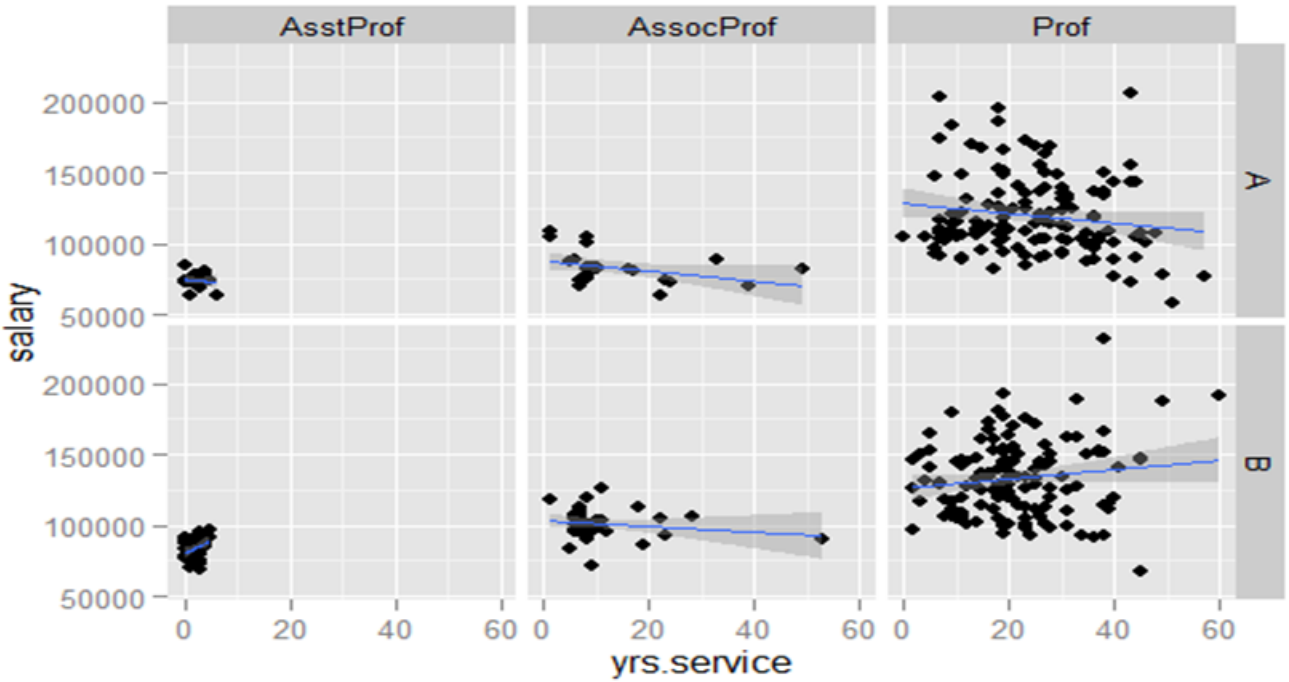
Facet Examples

```
p+geom_point()+geom_smooth(method="lm")+facet_grid(.~rank)
```



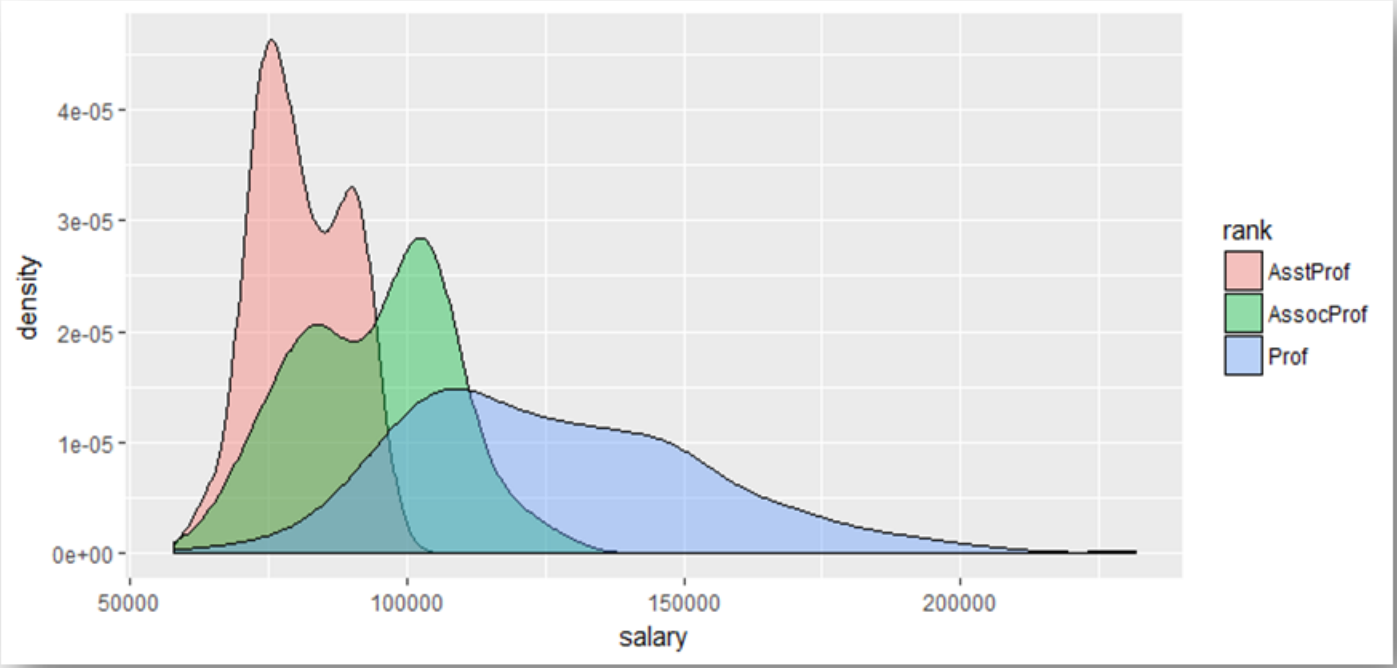
Facet Examples

```
p+geom_point()+geom_smooth(method="lm")+facet_grid(discipline~rank)
```



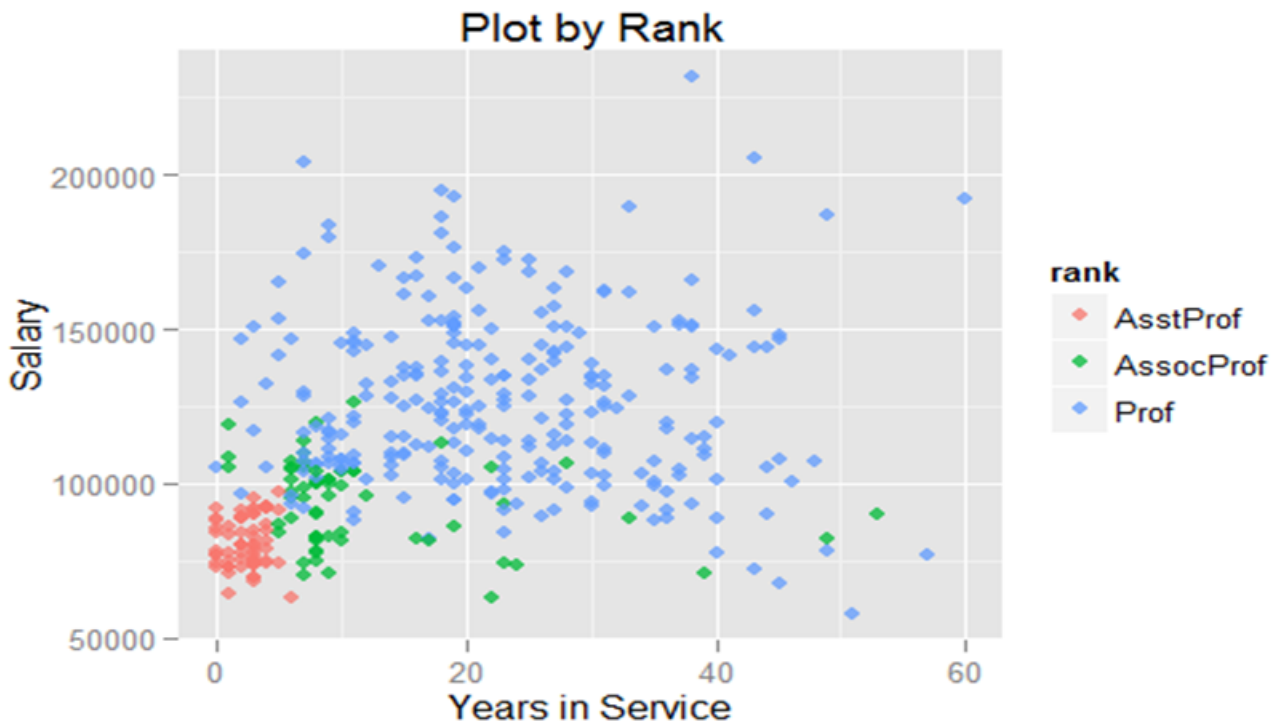
Density Plot

```
ggplot(Salaries, aes(x=salary,fill=rank))+geom_density(alpha=0.4)
```



Rendering labels

```
> p+geom_point(aes(color=rank), size=2, alpha=3/4)+labs(title="Plot by Rank")
+labs(x="Years in Service",y="Salary")
```



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Thank You